

Water Resources Research[®]

RESEARCH ARTICLE

10.1029/2024WR038353

Widespread Increasing Control of Water Supply on Evapotranspiration

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Key Points:

- The controls of water supply (soil moisture, SM) and water demand (vapor pressure deficit, VPD) on evapotranspiration (ET) are disentangled
- A significant increase in the effect of SM on ET is observed in 43% of water supply-limited regions
- Earth System Models cannot capture the trends of SM and VPD effects on ET

Supporting Information:

Supporting Information may be found in the online version of this article.

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Citation:

Zhang, Y., Liu, X., Wang, K., Zhang, D., & Liu, W. (2024). Widespread increasing control of water supply on evapotranspiration. *Water Resources Research*, 60, e2024WR038353. <https://doi.org/10.1029/2024WR038353>

Received 5 JUL 2024
Accepted 25 NOV 2024

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Funding acquisition: Xiaomang Liu
Investigation: Yu Zhang, Xiaomang Liu, Kaiwen Wang, Weihang Liu
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Abstract Evapotranspiration (ET), a crucial component of water consumption in the hydrological process, is directly controlled by soil moisture (SM) and vapor pressure deficit (VPD) from the perspectives of water supply and demand. However, SM and VPD are strongly coupled through multiple physical processes, confounding their effects on ET. Here, we decouple the interaction between SM and VPD and then analyze the spatiotemporal pattern of their individual effects on ET based on multiple observation-based data sets. The results show that ET is limited by SM rather than VPD in approximately 63% of global land areas (60°S–60°N), defined as water supply-limited regions. From 1982 to 2014, the effect of SM on ET enhances significantly in 43% of the water supply-limited regions. The trends can be attributed to changes in SM and VPD themselves as well as to changes in vegetation conditions. Using the findings from the observation-based data sets as the benchmark, we show that Earth System Models (ESMs) can overall reproduce the spatial pattern of SM and VPD effects on ET but fail to capture their temporal trends. Our results highlight that the water supply and demand control on ET varies with changing environments, which should be explicitly considered when analyzing the terrestrial water cycle and land-atmosphere interaction.

Plain Language Summary Evapotranspiration (ET) plays a crucial role in the global water cycle and is directly affected by water supply (e.g., soil moisture, SM) and atmospheric evaporative demand (e.g., vapor pressure deficit, VPD). In this study, we disentangle the effects of SM and VPD on ET and analyze their spatiotemporal patterns using multiple observation-based data and Earth System Models (ESMs). Our findings suggest that the significant increasing trend in SM control on ET is widespread in water supply-limited regions from 1982 to 2014. However, ESMs can not reproduce the trends derived from observation-based data. Our research enhances the understanding of how ET responds to water supply and demand in a changing environment, aiding in predicting water resource availability. This work also underscores the need to improve the capability of ESMs to better simulate the dynamic response of ET to its influencing factors.

1. Introduction

Evapotranspiration (ET), the second largest hydrological element after precipitation, returns approximately two-thirds of the precipitation falling on land back to the atmosphere, affecting the water cycle profoundly (Dorigo et al., 2021; Elnashar et al., 2021; Trenberth et al., 2007). As a crucial process connecting land and atmosphere, ET is co-regulated by water supply and atmospheric evaporative demand (Budyko & Miller, 1974; Jung et al., 2010; Li, Wang, et al., 2022). Soil moisture (SM) typically serves as the primary water supply for both soil evaporation and vegetation transpiration which constitute the majority of terrestrial ET (Jung et al., 2010; Purdy et al., 2018; Zhang et al., 2017). Meanwhile, vapor pressure deficit (VPD) is considered a concise and effective proxy for atmospheric evaporative demand (Novick et al., 2016; Peng et al., 2018). Thus, clarifying the effects of SM and VPD on ET from a water supply and demand perspective has important implications for estimating water availability and understanding the feedback between climate and water cycle (Berg & Sheffield, 2019; Li, Reichstein, et al., 2023; Zhou et al., 2021).

Generally, the effects of SM and VPD on ET are modulated by climate dryness (Novick et al., 2016). SM plays a crucial constraining role in ET, especially under dry conditions (Dong et al., 2022; Jung et al., 2010; Lian et al., 2021; Purdy et al., 2018). When water is abundantly available, VPD becomes more influential on ET (Vicente-Serrano et al., 2020). Several studies have investigated the effects of SM or VPD on ET based on observed or simulated data (Berg & Sheffield, 2018; Jung et al., 2010; Massmann et al., 2019). For instance, Jung

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et al. (2010) found that the increasing SM limitation accounts for the decline of global land ET from 1998 to 2008. The essential role of VPD in affecting different components of ET within the carbon–water–energy cycle has garnered increasing attention in recent years (Grossiord et al., 2020; Li, Wang, et al., 2022; Massmann et al., 2019; Zhang et al., 2021). However, SM and VPD are strongly coupled through large-scale atmospheric circulation and land-atmosphere interaction (Berg et al., 2015; Zhou et al., 2019), which tends to confound their individual effect on ET. Therefore, it is necessary to remove the dependence between SM and VPD before analyzing water supply and demand control on ET (Novick et al., 2016; Zhang, Koch, et al., 2023).

Recently, some studies have disentangled the relative effects of SM and VPD on vegetation index (e.g., gross primary productivity and normalized difference vegetation index) to understand how soil drought and atmospheric aridity influence ecosystem production (Fu et al., 2022; Liu et al., 2020; Novick et al., 2016; Zhang, Zhang, et al., 2023). Although ET highly depends on vegetation conditions (Zeng et al., 2018), the effects of SM and VPD on ET differ from that on vegetation due to variations in water use efficiency under changing environments (Kannenbergh et al., 2024; Zheng et al., 2023). Furthermore, unlike the vegetation index, ET at large spatial scales cannot be directly observed and is typically estimated by different models or algorithms (Morton, 1983; Talsma et al., 2018; Tang et al., 2024; Wang & Dickinson, 2012). Significant discrepancies among various ET products make it more challenging to analyze the relationship between ET and its influencing factors. To our knowledge, the control of SM and VPD on ET within the framework of the water supply and demand remains less understood. In particular, the dynamic evolution of this relationship is notable under the changing environment (Zhang, Liu, & Bai, 2018). Climate change, whether drying or wetting, can directly modify the ET response to SM and VPD. Additionally, vegetation variation may affect the sensitivity of ET to VPD by altering plant water regulation strategies (i.e., water-conservative or water-intensive) (Massmann et al., 2019; Yang et al., 2023). However, the dynamic analysis remains insufficient and needs further investigation.

In this study, we aim to analyze the spatial-temporal patterns of SM and VPD effects on ET using multiple observation-based data sets and Earth System Model (ESM)-based simulations. First, we decouple the correlation between SM and VPD to distinguish their respective effects on ET during the growing season. Then, we assess the spatial patterns of SM and VPD effects on ET and classify the globe into water supply-limited and demand-limited regions. Additionally, we analyze the temporal variations of SM and VPD effects on ET over recent decades and the potential influencing factors. Finally, we employ 20 ESMs from the Coupled Model Inter-comparison Project Phase 6 (CMIP6) to evaluate the performance of the state-of-the-art ESMs in simulating the SM and VPD effects on ET against observation-based results.

2. Data and Methodology

2.1. Data

2.1.1. Observation-Based Data

In this study, ET estimates are derived from three global observation-based products covering the period 1982–2014 to reduce data uncertainty. The first product, derived from FLUXNET eddy covariance measurements and upscaled to a global scale using machine learning methods (referred to as MFCW), is provided by Li, Xiao, et al. (2023). The second product is obtained from the Global Land Evaporation Amsterdam Model (GLEAM), which estimates ET using the Modified Priestley–Taylor algorithm (Martens et al., 2017). The third product, estimated by the Breathing Earth System Simulator (BESS) model, which integrates the essential physical and biochemical processes of the land surface, is provided by Li, Ryu, et al. (2023). These ET products have been largely validated or intercompared in previous studies (Li, Xiao, et al., 2023; Tang et al., 2024). The subsequent computation is conducted independently for each ET product and then averaged to reduce the data-related uncertainties.

SM data is obtained from the land component of the fifth generation of European ReAnalysis (ERA5-Land) (Muñoz-Sabater et al., 2021). The ERA5-derived SM data has been validated against in situ observations and demonstrates reasonable skill (Li et al., 2020). As a state-of-the-art reanalysis data set, ERA5-Land offers several improvements in land simulation and has been well used to investigate the interaction among land and atmospheric variables (Cheng et al., 2022; Li, Migliavacca, et al., 2022). ERA5-Land provides SM for four depths, including 0–7 cm, 7–28 cm, 28–100 cm, and 100–289 cm. Here, the root zone is defined as the 0–100 cm layer, which can cover the majority of vegetation root depth of the global area (Luo et al., 2023; Yang et al., 2016).

Table 1
Summary of All Observation-Based Data Sets in This Study

Variable	Data set	Resolution		Time span	References
		Spatial	Temporal		
ET	MFCW	0.5°	Monthly	1982–2016	Li, Xiao, et al. (2023)
	GLEAM	0.25°	Monthly	1980–2022	Martens et al. (2017)
	BESS v2.0	0.05°	Monthly	1982–2019	Li, Ryu, et al. (2023)
SM	ERA5-Land	0.1°	Monthly	1950–present	Muñoz-Sabater et al. (2021)
T _a , AVP, PRE, and PET	CRU	0.5°	Monthly	1901–2022	Harris et al. (2020)
LAI	GIMMS_LAI4g	1/12°	Half-monthly	1982–2020	Cao et al. (2023)

Note. ET, evapotranspiration; SM, soil moisture; T_a, air temperature; AVP, actual vapor pressure; PRE, precipitation; PET, potential evapotranspiration; LAI, leaf area index.

Meteorological data, including air temperature, actual vapor pressure, precipitation, and potential evapotranspiration (PET), are available from the Climate Research Unit (CRU) data set. The leaf area index (LAI), a commonly used vegetation growth index, is obtained from the GIMMS_LAI4g data set (Cao et al., 2023). Detailed information about observation-based data sets is provided in Table 1. In this study, we focus on the growing season and only retain the months with the climatology temperature above 5°C (Körner et al., 2023).

2.1.2. Earth System Model (ESM) Simulations

To evaluate the performance of ESMs in simulating the effects of SM and VPD on ET, we select 20 ESMs from the historical simulations of CMIP6. We use a single realization from each model and construct the multi-model ensemble by applying equal weights to all models. Specific variables include ET, air temperature, relative humidity, SM content, and LAI on a monthly scale. Detailed information about the 20 ESMs is provided in Table S1 in Supporting Information S1.

In this study, all data sets (both observation-based data and ESM simulations) are resampled to a comparable spatial resolution of 0.5° × 0.5° through the bilinear interpolation method. Only land areas between 60°S and 60°N are considered, with non-vegetated areas masked out.

2.2. Methodology

2.2.1. Calculation of Vapor Pressure Deficit (VPD) and Aridity Index (AI)

VPD is calculated as the difference between saturated vapor pressure and actual vapor pressure. CRU climate data set provides the actual vapor pressure. For CMIP6, the actual vapor pressure is derived from the product of relative humidity and saturated vapor pressure, as shown in Equations 1–4:

$$VPD = SVP - AVP \quad (1)$$

$$AVP = \frac{RH}{100} \times SVP \quad (2)$$

$$SVP = 6.112 \times \left(1 + 7 \times 10^{-4} + 3.46 \times 10^{-6} \times P_{mst}\right) \times e^{\frac{17.67T_a}{T_a + 243.5}} \quad (3)$$

$$P_{mst} = 1013.25 \times \left(\frac{T_a + 273.16}{T_a + 273.16 + 0.0065 \times Z}\right)^{5.625} \quad (4)$$

where SVP and AVP are saturated vapor pressure and actual vapor pressure (hPa), respectively. *RH* is the relative humidity (%), *T_a* is the air temperature (°C), *P_{mst}* is the air pressure (hPa), and *Z* is the altitude (m).

Aridity index (AI), a widely used metric for dividing climate regimes, is calculated as the ratio of mean annual precipitation to mean annual PET (Middleton & Thomas, 1997). Regions with AI ≤ 0.65 are classified as drylands, while the remaining regions are defined as humid lands (Huang et al., 2017; Hulme et al., 1992).

2.2.2. Contribution Analysis

To disentangle the relative effects of SM and VPD on ET, we employ the percentile binning method proposed by Liu et al. (2020). This method has been widely used to decouple the correlation between SM and VPD at different time scales (Cheng et al., 2022; Lu et al., 2022; Nie et al., 2021; Qiu et al., 2023; Zhang, Zhang, et al., 2023). The underlying hypothesis is that SM and VPD are independent in short intervals of SM or VPD. In this case, both monthly SM and VPD are sorted into 10 bins (i.e., 0–10th, 10–20th, ..., and 90–100th percentile) for each grid. Then, the data in each VPD (or SM) bin are further sorted into 10 bins according to the value of SM (or VPD). The individual effect of SM (or VPD) on ET is quantified by aggregating the differences in ET between the maximum and minimum SM (or VPD) bins within each VPD (or SM) bin. This method can overcome the limitation that the ET response to SM and VPD may be non-linear by binning the data according to percentile. Here, the time series of ET in each grid has been standardized by Z-score transformations (i.e., subtracting their mean and dividing by their standard deviation) for spatial comparison. $\Delta ET(SM|VPD)$ and $\Delta ET(VPD|SM)$ are used to represent the relative effects of SM and VPD on ET after excluding the interaction between SM and VPD, which are calculated for each grid according to Equations 5 and 6:

$$\Delta ET(SM|VPD) = \frac{1}{M} \sum_{i=1}^M (ET_{i,p_{i,max}} - ET_{i,p_{i,min}}) \quad (5)$$

$$\Delta ET(VPD|SM) = \frac{1}{N} \sum_{j=1}^N (ET_{j,q_{j,max}} - ET_{j,q_{j,min}}) \quad (6)$$

where M and N is the number of populated SM and VPD bins (here $M, N = 10$), i and j are the specific number of VPD and SM bins, respectively. $p_{i,max}$ and $p_{i,min}$ are the maximum and minimum SM bins at the i th VPD bins, and $q_{j,max}$ and $q_{j,min}$ are the maximum and minimum VPD bins at the j th SM bins. Considering limited samples for the growing seasons in each grid, a 7×7 spatial moving window for each grid is used to increase samples (Wang et al., 2024; Yao et al., 2023). A positive value of $\Delta ET(SM|VPD)$ or $\Delta ET(VPD|SM)$ indicates a positive effect of SM or VPD on ET and vice versa.

In this study, we define soil water supply-limited regions (hereinafter referred to as water supply-limited regions) for which $\Delta ET(SM|VPD) > \Delta ET(VPD|SM)$ and define atmospheric water demand-limited regions (hereinafter referred to as water demand-limited regions) for which $\Delta ET(VPD|SM) > \Delta ET(SM|VPD)$.

2.2.3. Trend Analysis and Drivers of the Trends

An 11-year moving window from 1982 to 2014 (i.e., 1982–1992, 1983–1993, ..., and 2004–2014) is used to explore the dynamic evolution of SM and VPD effects on ET. The effects of SM and VPD on ET (i.e., $\Delta ET(SM|VPD)$ and $\Delta ET(VPD|SM)$) are computed repeatedly for each temporal window, generating a time series of the effects. The nonparametric Mann-Kendall (M-K) test is applied to detect the trends of these effects, with Sen's slope estimator determining the magnitude of the trends. Meanwhile, the partial Mann-Kendall (pM-K) test is further employed to identify the possible driver of the significant trends. The pM-K test modifies the M-K test by removing the contribution of a given covariate (i.e., potential drivers) that correlates with the response variable. If the original significant trend of the response variable becomes non-significant after removing the influences of any covariate through the pM-K test, it can be inferred that the covariate significantly affects the detected trend in the response variable (Jain et al., 2022). Otherwise, there may be other potential drivers that could explain the trend but are not included. The contribution of each factor is then quantified by the difference between the M-K test and pM-K test statistics (Ma & Yuan, 2024; Yao et al., 2016), as shown in Equation 7:

$$CR_i = \frac{Z_{mk} - Z_{pmk,i}}{\sum_{i=1}^M (Z_{mk} - Z_{pmk,i})} \quad (7)$$

where Z_{mk} is the Z value from the M-K test, M is the number of considered explanatory factors, and $Z_{pmk,i}$ is the Z value from the pM-K test with the i th factor. The factor with the largest relative contribution is identified as the dominant driver for the detected significant trend.

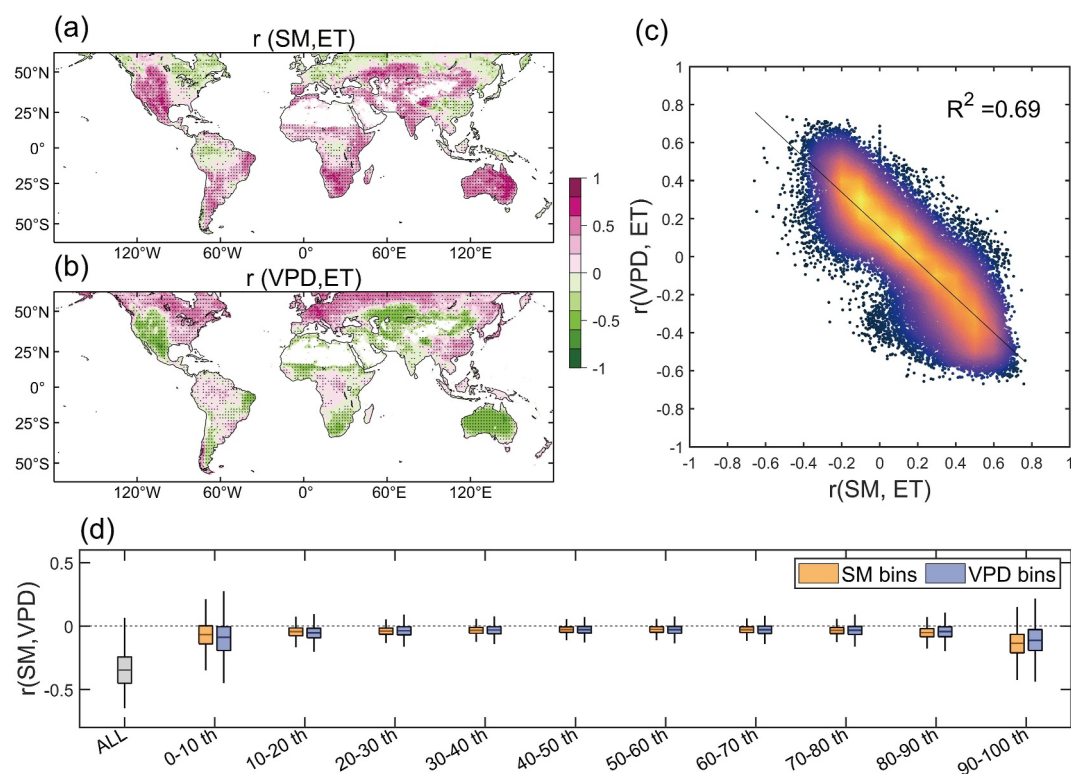


Figure 1. The effects of SM and VPD on ET are intertwined during the growing season. The spatial distributions of the correlation coefficient between (a) SM and ET ($r(\text{SM}, \text{ET})$) and that between (b) VPD and ET ($r(\text{VPD}, \text{ET})$). The results are averaged across the correlation coefficient computed from three ET products. Stippling indicates that at least two out of three products show statistically significant correlations ($p\text{-value} < 0.01$). (c) The relationship between $r(\text{SM}, \text{ET})$ and $r(\text{VPD}, \text{ET})$. (d) Boxplots of the correlation coefficients between SM and VPD (in gray) and those decoupled in each SM and VPD bin (in orange and blue) across all land grids.

Here, the trends in SM, VPD, and LAI are performed as possible drivers to explain the trends in SM and VPD effects on ET. SM and VPD, typically used as proxies for land and atmospheric dryness (Fu et al., 2022), directly influence their effects on ET. Although SM and VPD are initially decoupled to calculate their individual effects on ET within each time window, trends in either variable still shape the variation in the effect of the other on ET. LAI, as a representative indicator of vegetation structure and functioning, modulates the partitioning of available water into ET (Liu et al., 2020; Yang et al., 2023). These drivers are averaged over each spatial and temporal window corresponding to the time series of $\Delta\text{ET}(\text{SM}|\text{VPD})$ and $\Delta\text{ET}(\text{VPD}|\text{SM})$. The computation procedure for the M-K and pM-K tests is implemented using the R package “trend.”

3. Results

3.1. Decoupling the Dependence Between SM and VPD Across Global Land

The global patterns of the relationships among SM, VPD, and ET during the growing season are shown in Figure 1. The correlations between SM and ET and that between VPD and ET exhibit almost opposite spatial patterns (Figures 1a and 1b). This is due to the widespread significant negative correlation between SM and VPD across global land (Figure S1 in Supporting Information S1), intertwining their effects on ET (Figure 1c). For instance, a negative SM-ET relationship in high latitudes and some regions of low latitudes can be related to VPD-driven ET, which in turn consumes SM. Similarly, the negative VPD-ET relationship in mid-latitudes can be partly attributed to high VPD tending to be accompanied by low SM, thereby limiting ET. This suggests that the strong covariability between SM and VPD may interfere with our understanding of water supply and demand control on ET. Therefore, we use the percentile binning method to decouple SM and VPD before interpreting their effects on ET. The correlations between SM and VPD within each bin are rather lower and closer to zero with

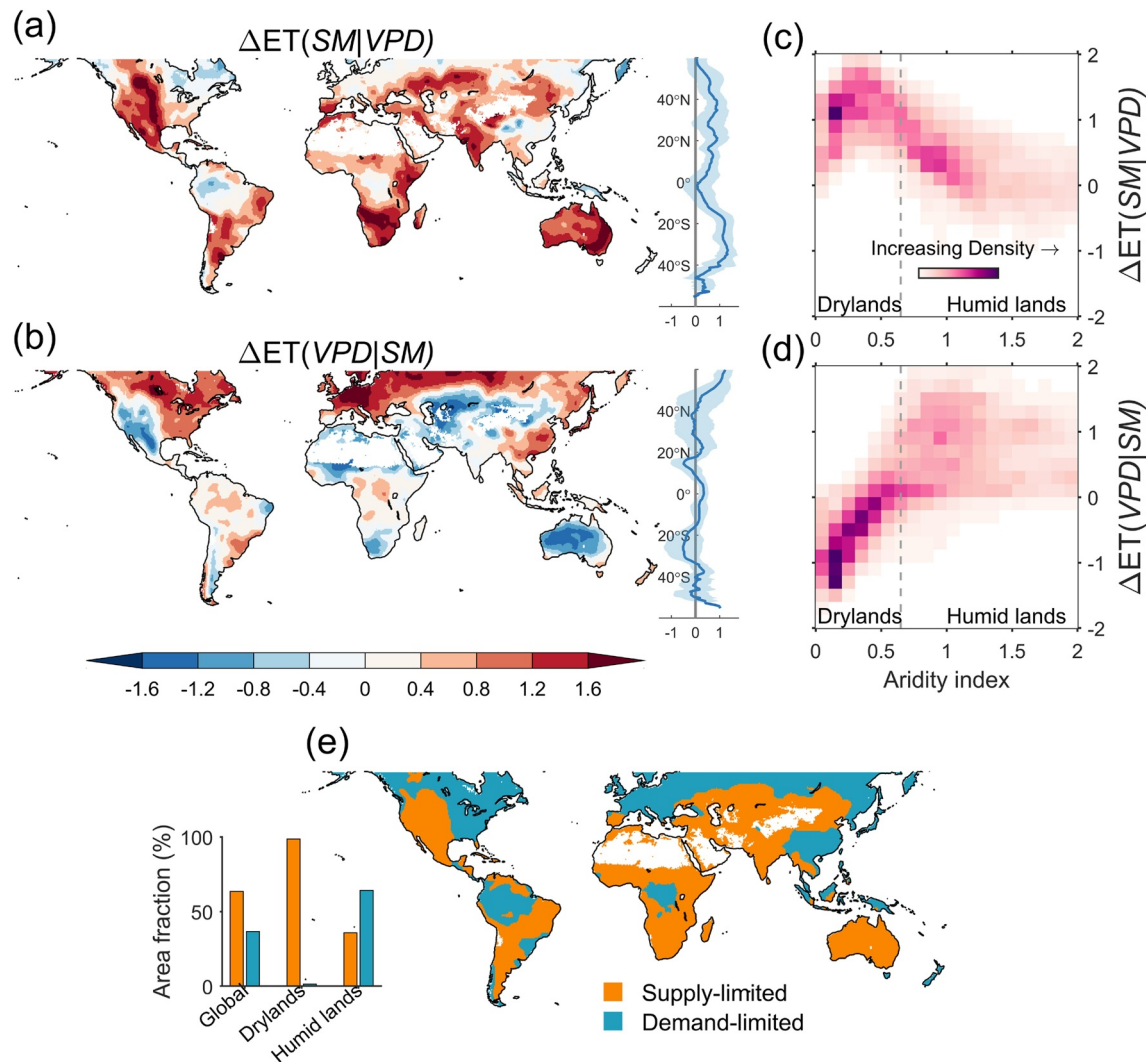


Figure 2. Relative effects of SM and VPD on ET globally. (a, b) Spatial distribution of $\Delta ET(SM|VPD)$ and $\Delta ET(VPD|SM)$ with their latitudinal distribution in the right portion of the panel. The solid line indicates the mean value at the same latitude, with shading representing one standard deviation. (c, d) $\Delta ET(SM|VPD)$ and $\Delta ET(VPD|SM)$ across all land grids grouped by the aridity index. (e) Spatial distribution of water supply-limited and demand-limited regions with their area fraction across global, drylands, and humid lands after area-weighted averaging.

respect to that across the entire time series (Figure 1d), indicating that SM and VPD can be decoupled based on this method.

3.2. Disentangling the Effects of SM and VPD on ET Across Climate Regimes

Figure 2 shows the spatial distribution of the relative effects of SM and VPD on ET. SM has a positive effect on ET in most regions, intuitively reflecting that more water supply promotes ET. A strong effect of SM on ET is observed in western North America, India, South Africa, and Australia (Figure 2a). VPD has a positive effect on ET in relatively humid regions, including high latitudes, the tropics, and parts of mid-latitudes (Figure 2b). Meanwhile, large regions of mid-latitudes show a negative VPD effect on ET, attributed to the fact that although higher VPD demands more moisture from the soil but also prompts vegetation to close stomata to reduce water loss (Li, Wang, et al., 2022; Zhong et al., 2023). The spatial distribution of SM and VPD effects on ET shows a clear dependence on climate regime. As climate shifts from wet to dry (i.e., as AI decreases), the effect of SM on ET initially strengthens and then weakens, while the effect of VPD on ET transitions from positive to negative (Figures 2c and 2d). This is consistent with the fact that water demand control on ET strengthens as the water

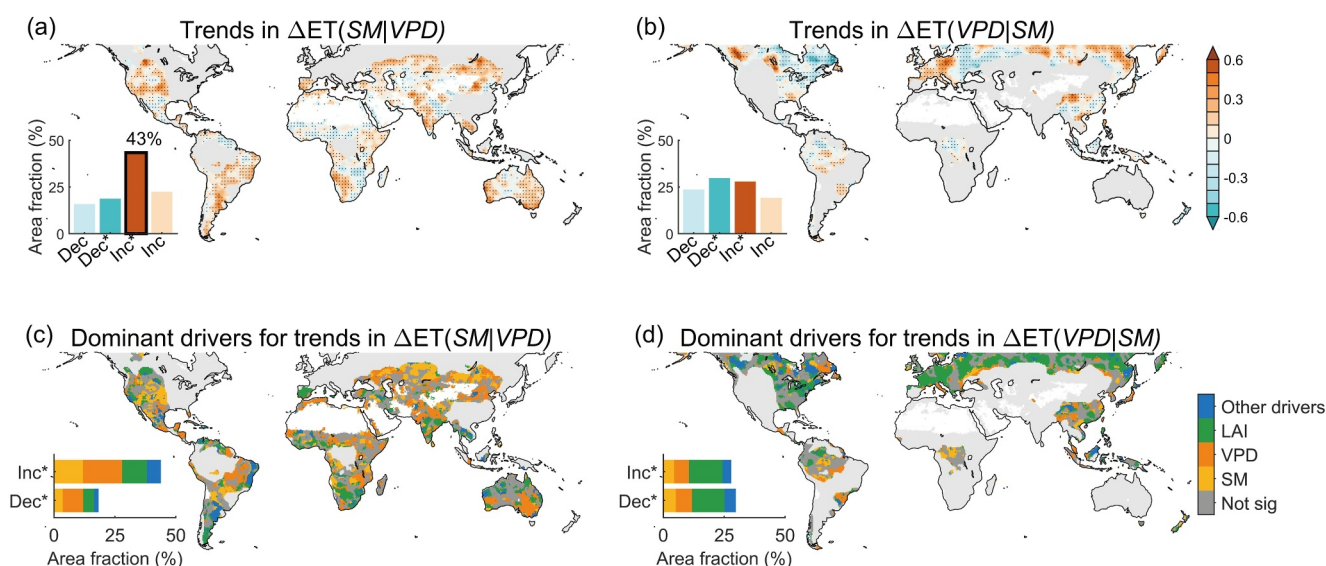


Figure 3. Global patterns of trends in SM and VPD effects on ET and their dominant drivers during 1982–2014. (a, b) Spatial distribution of trends (Sen's slope, per decade) in $\Delta ET(SM|VPD)$ and $\Delta ET(VPD|SM)$ across water supply-limited and demand-limited regions, respectively. Stippling indicates that the trend is statistically significant (M-K test, p -value < 0.1). Area fractions with significantly and non-significantly increased (Inc* and Inc) and decreased (Dec* and Dec) trends are shown at the bottom left of each panel. (c, d) Dominant drivers for trends in $\Delta ET(SM|VPD)$ and $\Delta ET(VPD|SM)$ using the attribution test based on the pM-K method. Regions without significant trends are masked in dark gray. Area fractions with significant trends dominated by each driver are shown at the bottom left of each panel.

supply is sufficient (Seneviratne et al., 2010). The low effect of SM on ET in arid and hyper-arid regions (i.e., $AI < 0.20$) may be related to the low fluctuation of SM.

Globally, ET is primarily limited by SM in 63% of land areas (defined as water supply-limited regions), encompassing almost all drylands (Figure 2e and Figure S2 in Supporting Information S1). The remaining regions where ET is limited by VPD (defined as water demand-limited regions) are mainly located in humid regions, including high latitudes, the Amazon rainforest, central Africa, Southeast Asia, and southern China. We note that the classification of water supply-limited and demand-limited regions is based on the dominant factor influencing ET from the climatology and does not imply that ET in these regions is unaffected by another factor. These regimes may also shift under extreme climate conditions. We also use the partial correlation method to test the results based on the percentile binning approach, which shows a similar spatial pattern as shown in Figure S3 in Supporting Information S1.

3.3. Increasing SM Effect on ET in Water Supply-Limited Regions

The dynamic evolution of SM and VPD effects on ET is shown in Figure 3. We focus on the variation in the effect of SM on ET in water supply-limited regions and the effect of VPD on ET in water demand-limited regions. In water supply limited regions, 43% exhibit significant increasing trends in SM effect on ET (p -value < 0.1), while only 19% show significant decreasing trends. The strengthening effect of SM on ET is notably widespread, primarily located in western North America, Australia, and large areas of Asia (Figure 3a). In contrast, the changes in the effect of VPD on ET in water demand-limited regions are relatively spatially heterogeneous, with areas showing significant decreases (30%) only slightly more than those showing significant increases (28%; Figure 3b). The weakening effect of VPD on ET is pronounced in regions such as eastern North America and eastern Europe.

Then, we analyze the cause of trends in SM and VPD effects on ET, attributing these significant trends to SM, VPD, and LAI trends. Figures 3c and 3d show the distribution of dominant factors contributing to the significant trends in SM effect on ET and VPD effect on ET, respectively (Figure S4 in Supporting Information S1 shows the relative contribution of each driver averaged over corresponding regions). In the water supply-limited regions, the strengthening SM effect on ET is not only determined by SM drying but also by changes in VPD in many areas (Figure 3c and Figure S5 in Supporting Information S1). For example, in western North America and central Asia, the enhanced SM effect on ET is primarily driven by soil drying, whereas in eastern Asia and eastern Australia, it

is more likely linked to atmospheric drying. This is because, as atmospheric water demand increases induced by warming, ET consumes more available water from SM when SM is high. On the other hand, plants try to conserve water by closing stomata in response to increased atmospheric water stress, resulting in a more pronounced decline in ET when SM is low. In the water demand-limited regions, the variation in the VPD effect on ET is more largely related to vegetation changes, particularly in the high latitudes (Figure 3d). The vegetation greening (e.g., increasing foliage cover) tends to enhance the proportion of available water being partitioned into ET (Yang et al., 2023; Zhang, Yang, et al., 2018), making ET more sensitive to VPD. Overall, our findings highlight that the increasing SM effect on ET is widespread in water supply-limited regions over the past several decades, which can mainly be attributed to soil and atmospheric drying.

3.4. ESMs Hardly Capture the Increasing SM Effect on ET

To evaluate the performance of ESMs in simulating the SM and VPD effects on ET, we further compare the observation-based results with simulations from 20 CMIP6 ESM outputs. ESMs can overall reproduce the spatial distribution of SM and VPD effects on ET and the division of water supply-limited and demand-limited regions (Figure S6 in Supporting Information S1). In most regions, the consistency among models is high, with their multi-model mean agreeing on the sign of observation-based results. However, in some tropical regions and dry–wet transition zones, ESMs show relatively low consistency among models of the VPD effect on ET. Approximately 75% of global land areas are divided into water supply limited regions through ESMs, higher than the observation-based result (63%). Note that multi-model means may smooth some signals, such as the magnitude of these effects varying with regions.

Although ESMs can roughly reproduce the spatial pattern of SM and VPD effects on ET, they fail to reoccur the trends of these effects obtained from observation-based data (Figure 4 and Figure S7 in Supporting Information S1). After averaging the results in water supply-limited regions, we find that ESMs show a significant decrease in SM effect on ET (-0.004 per decade; p -value <0.01), which is opposite to the significant increase (0.059 per decade; p -value <0.01) detected by observation-based data (Figure 4a). Similarly, after averaging the results in water demand-limited regions, ESMs overestimate the observation-based decreases in the VPD effect on ET (Figure 4b). Specifically, the former shows a significant decreasing trend (-0.022 per decade, p -value <0.01), but the latter shows a slight and not significant decreasing trend (-0.005 per decade).

The trends derived from individual models also reveal significant deviations from those derived from observation-based data sets, as demonstrated by their probability density functions (Figures 4c and 4d). Although the grid-scale trends are centered around zero for most individual models, they exhibit considerable spatial disparity across models (Figure S7 in Supporting Information S1). The ESM ensemble mean can capture the signs of observation-based trends in only about half of the areas (Figures 4e and 4f). Moreover, most individual models show substantial spatial inconsistency with observation-based trends (Figure S8 in Supporting Information S1).

The poor capability of ESMs to simulate the changes in SM and VPD effects on ET is related to their inaccuracy in simulating the influencing factors (i.e., SM, VPD, and LAI trends; Figures S5 and S9 in Supporting Information S1). ESMs can generally reproduce the observation-based trends in VPD, but they show inconsistent trends in SM and LAI across many regions. For example, ESMs fail to capture the increasing SM effect on ET in western North America and central Asia, which can be partly attributed to their inability to capture the SM drying trend. Similarly, ESMs cannot reproduce the increasing SM effect on ET in southern Africa, likely due to their inability to simulate the greening trend. Apart from these direct factors, the reasons for the poor performance of ESMs in simulating the changes in SM and VPD effects on ET are complex (e.g., CO_2 fertilization effect), which are beyond the scope of this study and not discussed further. Overall, our findings demonstrate that temporal variability of ET response to SM and VPD simulated by ESMs involves considerable uncertainty.

4. Discussion

4.1. Implications

Terrestrial ET is governed by the water supply on the land surface (e.g., SM) and water demand for the atmosphere (e.g., VPD) and determines the partitioning of water and energy fluxes between the land and atmosphere (Trenberth et al., 2007). However, the responses of ET to SM and VPD are not independent but rather intertwined. For instance, SM deficit can reduce ET and potentially reduce atmospheric moisture. The decreased ET cooling

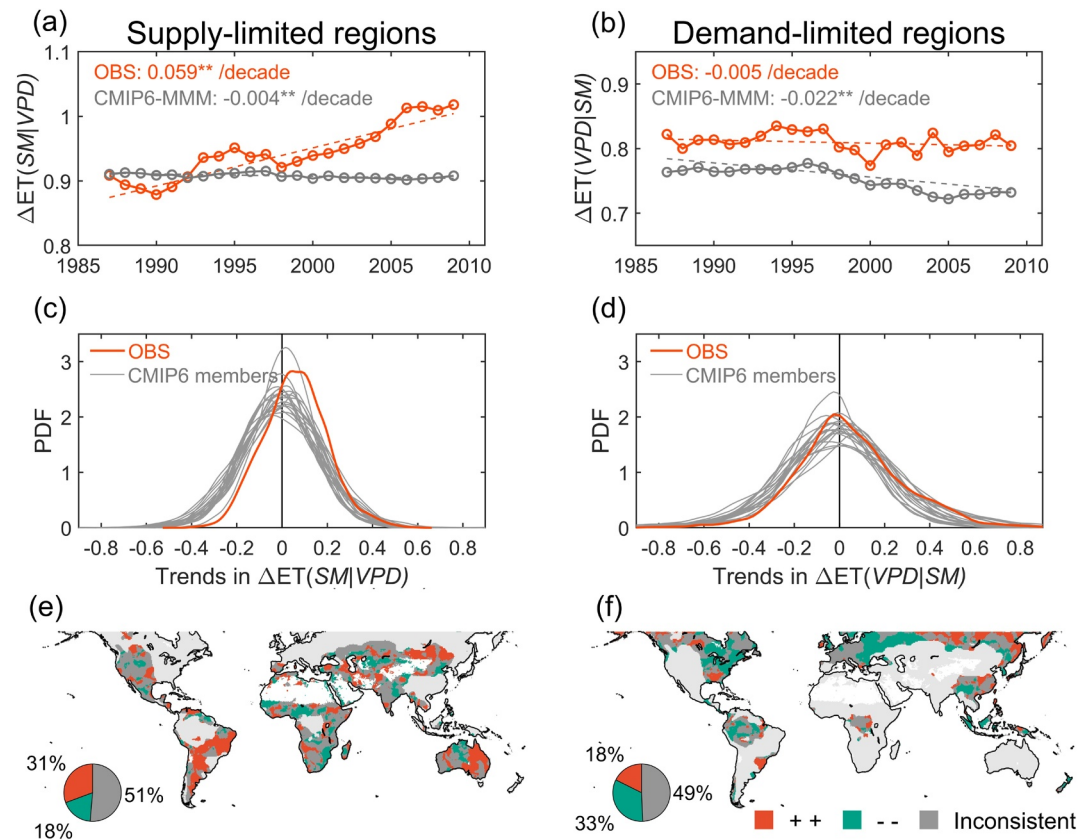


Figure 4. Comparison of observation-based and ESM-based trends in SM and VPD effects on ET. (a, b) Temporal variations of $\Delta ET(SM|VPD)$ and $\Delta ET(VPD|SM)$ are area-weighted averaged across water supply-limited regions and demand-limited regions, respectively. Orange and black lines represent the time series from observation-based data sets (OBS) and multi-model ensemble mean (MMM) of CMIP6. The slope values of trends are provided, with ** denoting significant trends at the 0.01 level, respectively. (c, d) Probability density functions (PDF) for trends across all grids estimated by nonparametric kernel density method. Gray lines indicate results from all ensemble members for CMIP6. (e, f) Global spatial consistency between ESM-based and observation-based trends. Red (++) indicates that both the ESM-based and observation-based trends are positive; green (--) indicates both are negative; and dark gray denotes inconsistent signals.

effect can increase sensible heat, resulting in higher surface air temperature. The combination of higher temperature and lower atmospheric moisture can enhance VPD. Higher VPD raises the atmospheric evaporative demand, accelerating ET and further depleting SM (Zhang et al., 2022; Zhou et al., 2019). This feedback or interaction can also explain the frequent co-occurrence of drought and heatwaves (Hao et al., 2022; Miralles et al., 2019).

In the context of global change, the response of ET to SM and VPD are likely to change over time. The changing response of ET to the environment is crucial for understanding shifts in the hydrological cycle, land-atmospheric coupling, and associated extreme events (Liu et al., 2021; Maurer et al., 2022). Our study observes a significantly increasing effect of SM on ET in recent decades, suggesting that the interaction between land and atmosphere becomes more active, potentially raising the risk of hydroclimatic extremes such as drought and heatwaves (Qing et al., 2023; Seo & Ha, 2022). This study also indicates that climate and vegetation variation can drive the changes in the SM and VPD effects on ET. We further speculate that the sensitivity of ET to water supply and demand is likely to shift in a warming future (Cook et al., 2014; Greve & Seneviratne, 2015).

ESMs are valuable tools for simulating and projecting the relationship between ET and other land and atmospheric variables (Berg & Sheffield, 2019; Hsu & Dirmeyer, 2023; Zhou et al., 2024). However, our study indicates that ESM simulations from the state-of-the-art CMIP6 fail to reproduce the changes in SM and VPD effects on ET and show significant inter-model uncertainties. The simulation errors in ESMs possibly arise from the oversimplification and incomplete treatment of key processes, along with inaccuracies in parameterization.

For instance, ESMs tend to use macroscopic approaches to simulate root water uptake while disregarding the soil–root hydraulic gradient, limiting the capability to simulate land–atmosphere fluxes (Clark et al., 2015; Xie et al., 2024). A previous study has suggested that the representation of soil evaporation stress is typically oversimplified in ESMs, which primarily contributes to the bias in the relationship between SM and ET (Dong et al., 2020). Additionally, the sensitivity of ET to vegetation change is highly uncertain in the current ESMs due to inaccuracies in plant parameterizations and groundwater processes (Wang & Zeng, 2024). Overall, ESMs face challenges in simulating the complex coupling between the hydrological cycle, vegetation, and climate. Our findings suggest that the poor performance of ESMs in simulating the time-varying relationship between ET and its influencing factors likely prevents them from reliably projecting water availability and land–atmospheric interaction in future climates.

4.2. Limitations

There are some limitations in this study. First, the uncertainties in observation-based data need to be considered cautiously, even though we use multiple data sets to help reduce these uncertainties. Due to insufficient direct observations, ET remains one of the most uncertain hydrological elements, with considerable disparities across different sources (Dorigo et al., 2021; Tang et al., 2024). Commonly used ET products rely on remote sensing-based climate data and are estimated through ET algorithms, models, or machine learning methods (Bai & Cai, 2023; Elnashar et al., 2021). The generation processes of such ET data employ climate variables, potentially resulting in an overestimation of the role of VPD (Kannenberg et al., 2024).

Second, this study does not distinguish the individual components of ET, namely soil evaporation, vegetation transpiration, and canopy interception. The response of these components to SM and VPD is different or even opposite. For instance, high VPD can increase direct soil evaporation while simultaneously reducing transpiration through stomatal closure to conserve water for plants (Massmann et al., 2019; Zhang et al., 2021). The effect of SM on ET can also be confounded by the increased relationship between precipitation and canopy interception in regions with enhanced precipitation and vegetation (Berg & Sheffield, 2019; Cui et al., 2022).

Finally, this study focuses on the direct driving factors of ET from the perspective of water supply and demand, with SM representing water supply and VPD serving as the proxy of atmospheric evaporative demand. However, other potential factors affecting ET are not directly considered, such as radiation, temperature, wind speed, CO₂ concentration, and land use change (Denissen et al., 2022; Liu et al., 2021; Yang et al., 2019, 2023). These factors are coupled tightly through multiple physical processes, making it difficult to disentangle their individual impact on ET based on limited observation. The combination of conceptual models, statistical methods, and fully coupled ESM experiments with reliable parameterization is needed to better understand these interactions (Wang & Zeng, 2024; Zhou et al., 2023).

5. Conclusions

The present study combines multiple observation-based and ESM-based data to investigate the global spatial-temporal pattern of ET response to water supply (SM) and water demand (VPD). Spatially, the effects of SM and VPD on ET vary with local climate regimes. ET is constrained by SM in 63% of the global land areas (60°S–60°N), which are classified as water supply-limited regions, while the remaining areas are water demand-limited regions. Temporally, a widespread increasing control of SM on ET is observed in water supply-limited regions over the period 1982–2014, with 43% of these areas experiencing significant increasing trends. While ESMs can overall reproduce the spatial distribution of SM and VPD effects on ET, they perform poorly in simulating the trends in these effects.

Clarifying the respective roles of SM and VPD on ET is crucial for understanding water partitioning between the land and atmosphere. Our findings reveal the extensive strengthening control of SM on ET, which has important implications for assessments of the terrestrial water cycle and water–climate interaction. This study also highlights that the application of ESMs in simulating the sensitivity of ET to land and atmospheric variables needs to be cautious. More efforts in collaboration across multiple disciplines are required to improve ESMs to better describe the ET process, especially for applications in a changing future.

Data Availability Statement

All data used in this study are publicly available at the following websites. MFCW (Li, Xiao, et al., 2023): <https://globalecology.unh.edu/data/MF-CW.html>; GLEAM (Martens et al., 2017): <https://www.gleam.eu/#downloads>; BESS v2.0 (Li, Ryu, et al., 2023): <https://www.environment.snu.ac.kr/bessv2>; CRU (Harris et al., 2020): https://crudata.uea.ac.uk/cru/data/hrg/cru_ts_4.07/; ERA5-Land (Muñoz-Sabater et al., 2021): <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land-monthly-means?tab=download>; GIMMS LAI4g (Cao et al., 2023): <https://doi.org/10.5281/zenodo.7649107>; and CMIP6 (Eyring et al., 2016): <https://esgf-node.ipsl.upmc.fr/projects/cmip6-ipsl/>. Codes used to perform the analyses are available on GitHub (<https://github.com/zhangyu0517/Global-Inc-WS-ET>).

Acknowledgments

This research was supported by the National Key Research and Development Program of China (No. 2023YFC3206600). We acknowledge the editors and three reviewers for their constructive comments and suggestions for improving this paper. We also thank Sifang Feng and Ying Yao for their assistance during this study.

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